EXPNO:3	CLASSIFICATION WITH DECISION TREES

AIM

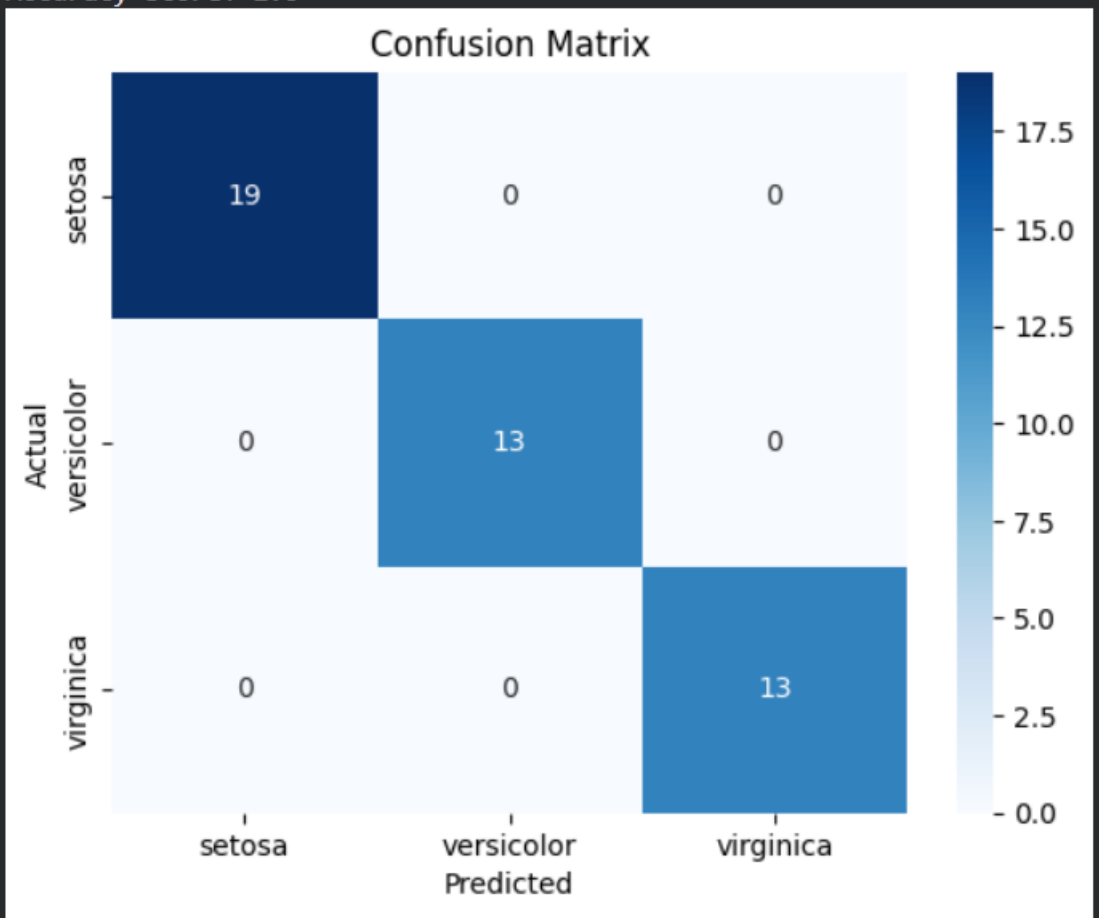
To implement a Decision Tree classifier and evaluate its performance using **accuracy score** and **confusion matrix** on a real-world dataset.

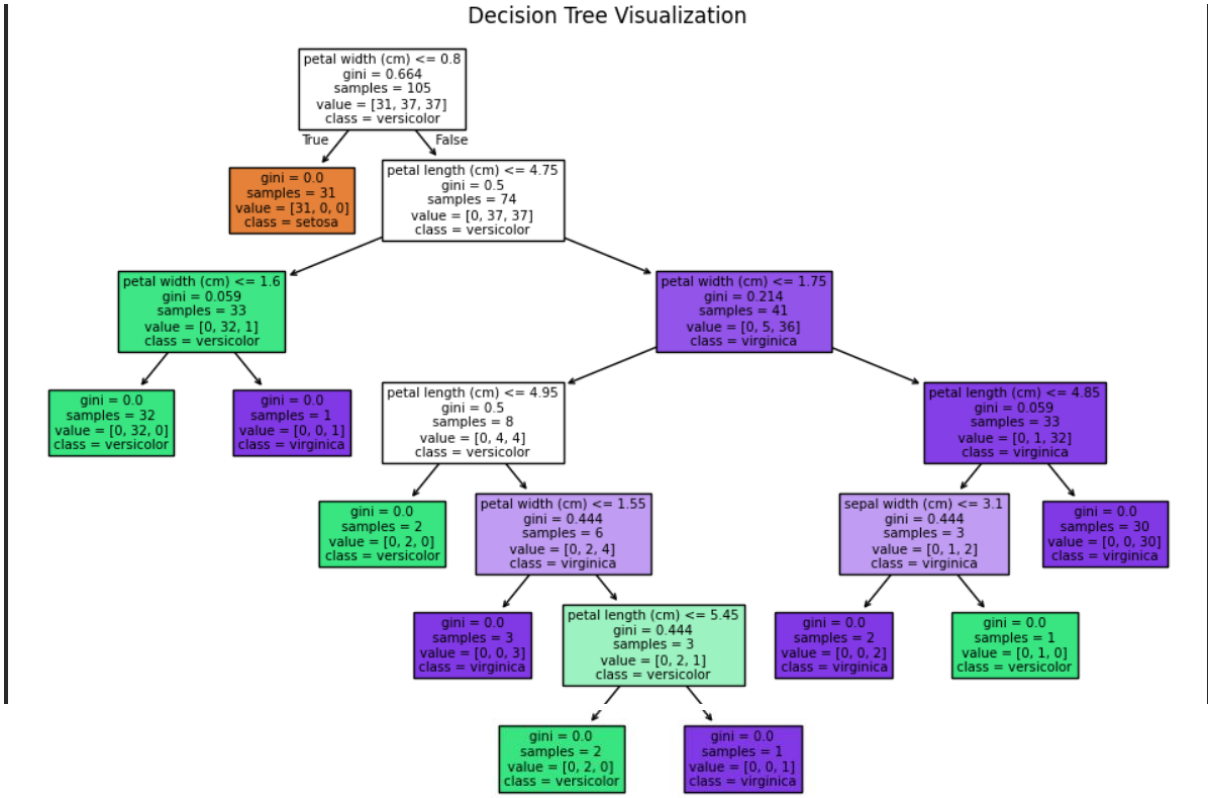
CODE1:

```
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix, accuracy_score
import matplotlib.pyplot as plt
import seaborn as sns

# Load Dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Train Decision Tree Classifier
dt_model = DecisionTreeClassifier(criterion='gini', random_state=0)
dt_model.fit(X_train, y_train)

# Predict
y_pred = dt_model.predict(X_test)

# Evaluate
cm = confusion_matrix(y_test, y_pred)
acc = accuracy_score(y_test, y_pred)
print("Confusion Matrix:\n", cm)
print("Accuracy Score:", acc)

# Visualize Confusion Matrix
sns.heatmap(cm, annot=True, cmap="Blues", xticklabels=iris.target_names, yticklabels=iris.target_names)
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()

# Visualize Decision Tree
plt.figure(figsize=(12, 8))
plot_tree(dt_model, filled=True, feature_names=iris.feature_names, class_names=iris.target_names)
plt.title("Decision Tree Visualization")
plt.show()
```

OUTPUT1:

```
Confusion Matrix:
 [[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
Accuracy Score: 1.0
```



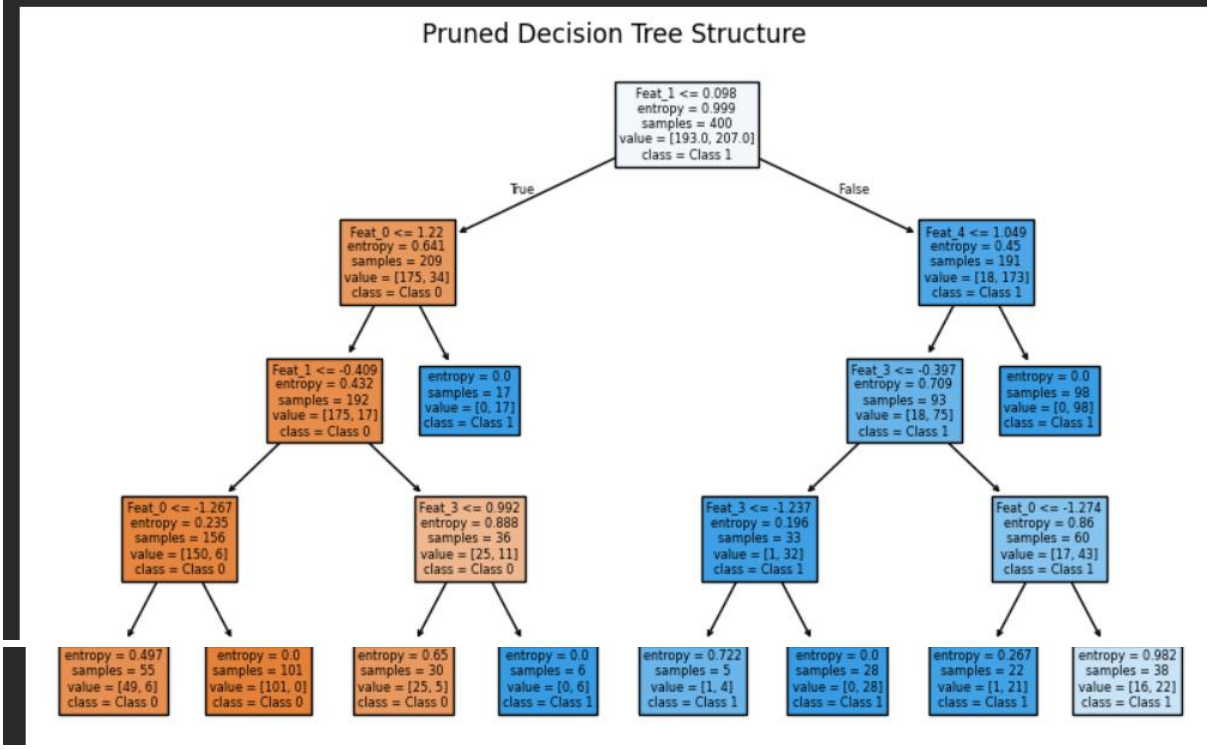


CODE2:

```
from sklearn.datasets import make_classification
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report
import matplotlib.pyplot as plt

# Generate Synthetic Classification Dataset
X, y = make_classification(n_samples=500, n_features=5, n_informative=3, n_redundant=1, random_state=42)

# Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train Decision Tree using Entropy and pruning parameters to prevent overfitting
dt_custom = DecisionTreeClassifier(criterion='entropy', max_depth=4, min_samples_split=10, random_state=42)
dt_custom.fit(X_train, y_train)

# Evaluate
y_pred = dt_custom.predict(X_test)
print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))

# Tree Visualization
plt.figure(figsize=(10, 6))
plot_tree(dt_custom, filled=True, feature_names=[f'Feat_{i}' for i in range(5)], class_names=['Class 0', 'Class 1'])
plt.title("Pruned Decision Tree Structure")
plt.show()
```

OUTPUT2:

```
Decision Tree Accuracy: 0.92
              precision    recall  f1-score   support

           0       0.92      0.95      0.93        58
           1       0.93      0.88      0.90        42

    accuracy                           0.92       100
   macro avg       0.92      0.91      0.92       100
weighted avg       0.92      0.92      0.92       100
```



RESULT:

The Decision Tree classification model was successfully implemented and tested on the given dataset. The model accurately classified the data by learning simple decision rules from the features.

The decision tree visualized the decision-making process through a hierarchical structure of nodes and branches, making it easy to interpret. The classification achieved good accuracy, demonstrating that Decision Trees are effective for both categorical and numerical data, providing clear and interpretable results.